

Technical test

Voicebot prototype for municipal services

Objective

Build a voicebot in Vapi that assists a citizen of the Municipality (e.g. Codroipo, or any other of your choice) in two use cases:

- answering questions about municipal services;
- simulating the booking of an appointment at a municipal office.

What to build

1. **Vapi Agent:** configured to manage the conversation and call tools when appropriate. Candidates may create their own Vapi account and use the free trial credits to test the agent.
2. **Backend with tools:** a small local project exposing the endpoints used by the agent, for example:
 - a retrieval tool for municipal service information (see below);
 - a tool to create/check an appointment (in-memory is fine).

The backend can run locally and be exposed to Vapi via ngrok.

Information retrieval (RAG)

3. Retrieve the content of Codroipo Municipality services via website scraping or any available public APIs.
4. Implement a simple retrieval mechanism, for example using open source embedding and similarity search tools. If hardware limitations apply, it is acceptable to pass hardcoded content to the AI and describe (or sketch) how a working RAG would be implemented.

What we evaluate

- ability to configure a Vapi agent that carries out a natural conversation in Italian;
- integration between the agent and external tools (local hosting + ngrok exposure);
- quality, clarity, and organisation of the backend code;
- reasoned choices (prompt design, tool design, edge case handling).

Deliverable

- repository with backend code and setup instructions;
- Vapi agent configuration/export (or instructions to recreate it);
- brief note on choices made, limitations, and what you would improve with more time;
- indicate any AI tools used and for what purpose.

Extra

Considered a plus:

- project containerised with Docker and orchestrated via docker-compose, with separate services for database, backend and frontend;
- small frontend to display booked appointments and/or conversation logs;
- appointment persistence on a database instead of in-memory.